Supplementary Information

Document caption: Detailed methods, supplementary analyses, and results including p-values.

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List of Figures

1 Current climate conditions

Data is available 2007 – 2022. 2008 is the first whole year of temperature data. Data available from https://data.g-e-m.dk (GEM 2026f, 2026d).

1-1 Temperature

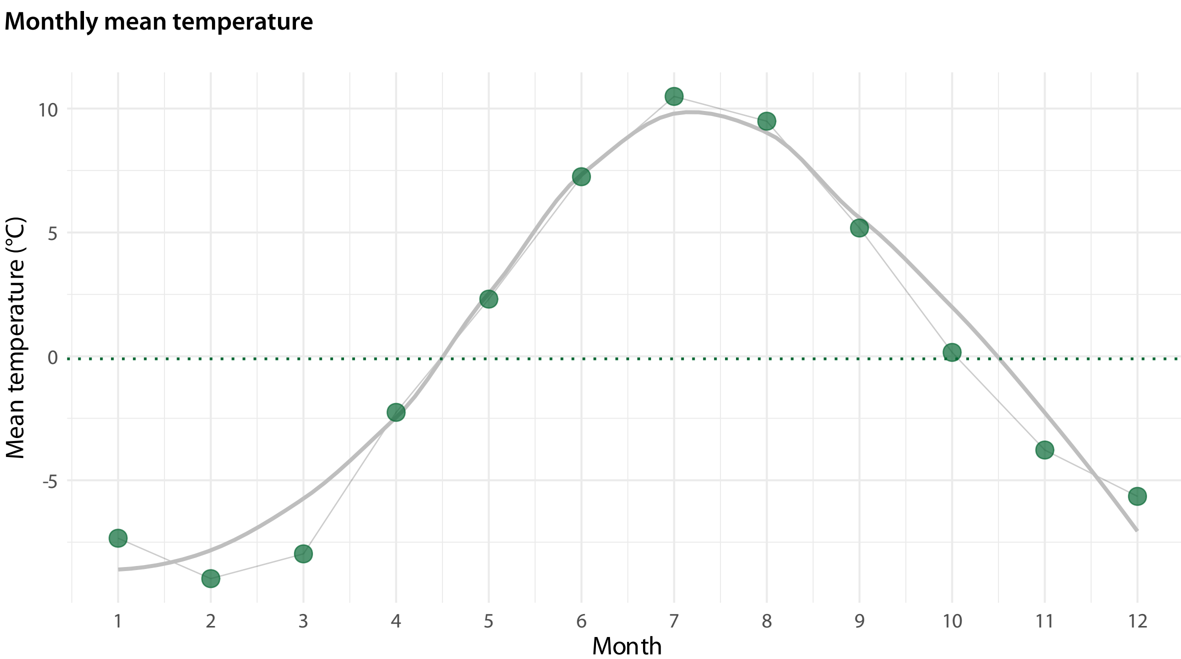

Figure 1: Mean temperature pr month. Mean air temperature (°C) pr month, based on data from 2007 to 2022. The dashed line represent yearly mean of -0.1°C.

The mean annual temperature is -0.1 °C. The warmest months are July (10.5 °C), August (9.5 °C), June (7.2 °C). The coldest months are January (-7.3 °C), March (-8 °C), February (-9 °C).

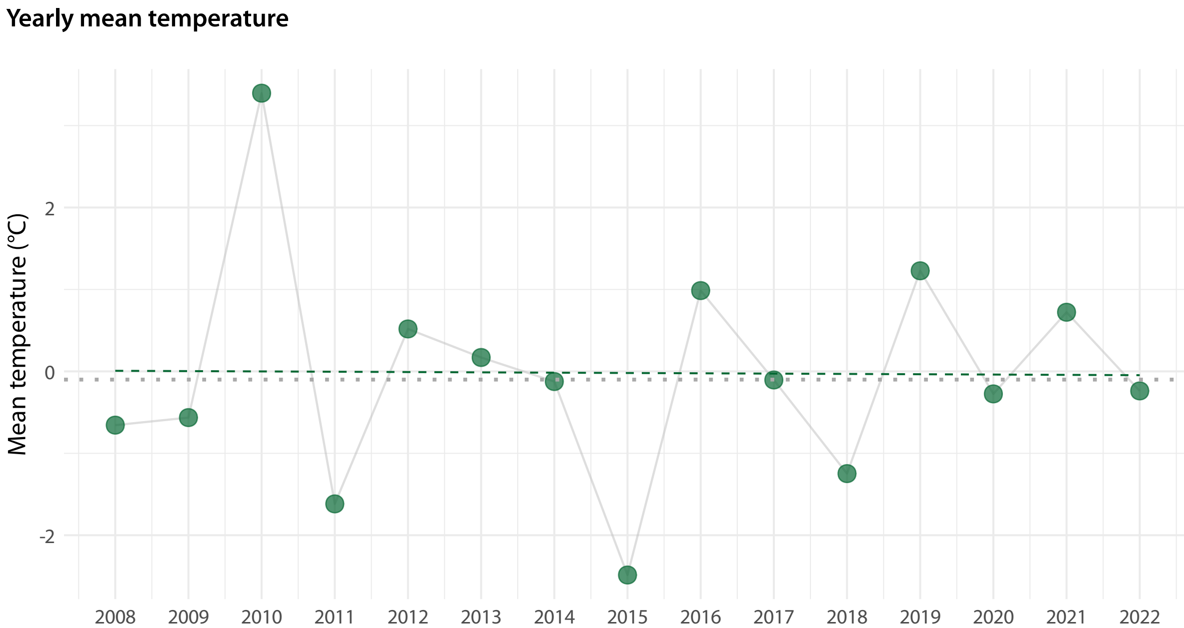

Figure 2: Mean annual air temperature pr year (°C) from 2008 to 2022 in Kangerluassunguaq. Data from 2007 is excluded because data is only from October, November and December. Gray dotted line indicate over all mean of -0.1003°C. Dashed line is model prediction.

Five year intervals of temperature

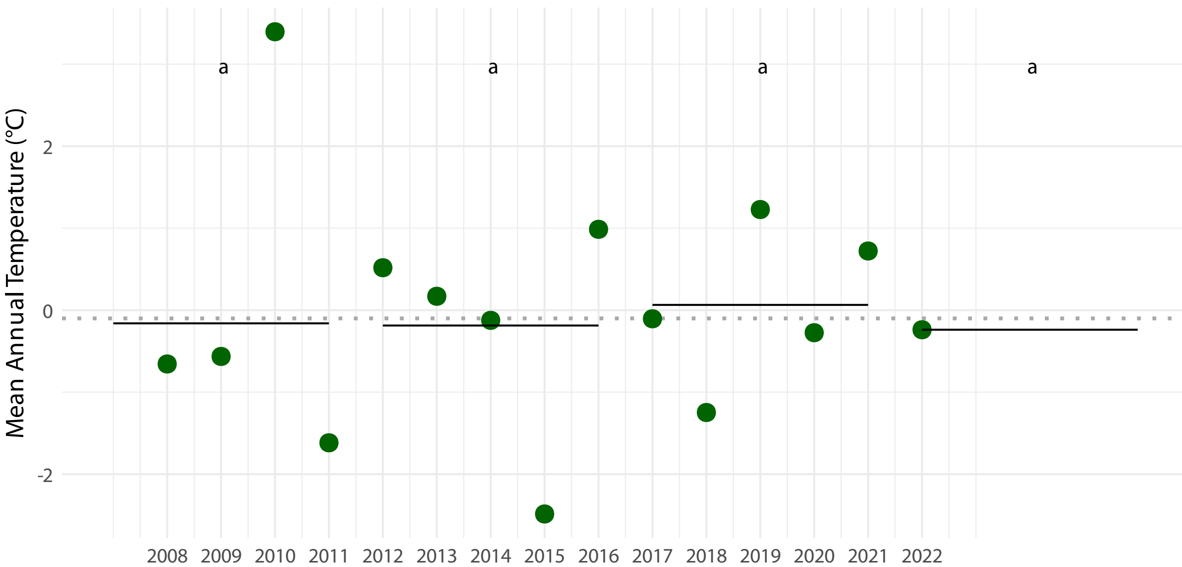

Figure 3: Five year intervals in temperature means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

1-2 Precipitation

Data available from https://data.g-e-m.dk (GEM 2026d).

The mean annual precipitation is 886.15 mm.

The months with the most precipitation are September (138.64 mm), October (99.38 mm), August (98.28 mm), which account for 336.3 mm or 38 % of the yearly precipitation.

The months with the least precipitation are May (51.51 mm), February (37.62 mm), March (36.44 mm).

Monthly mean precipitation

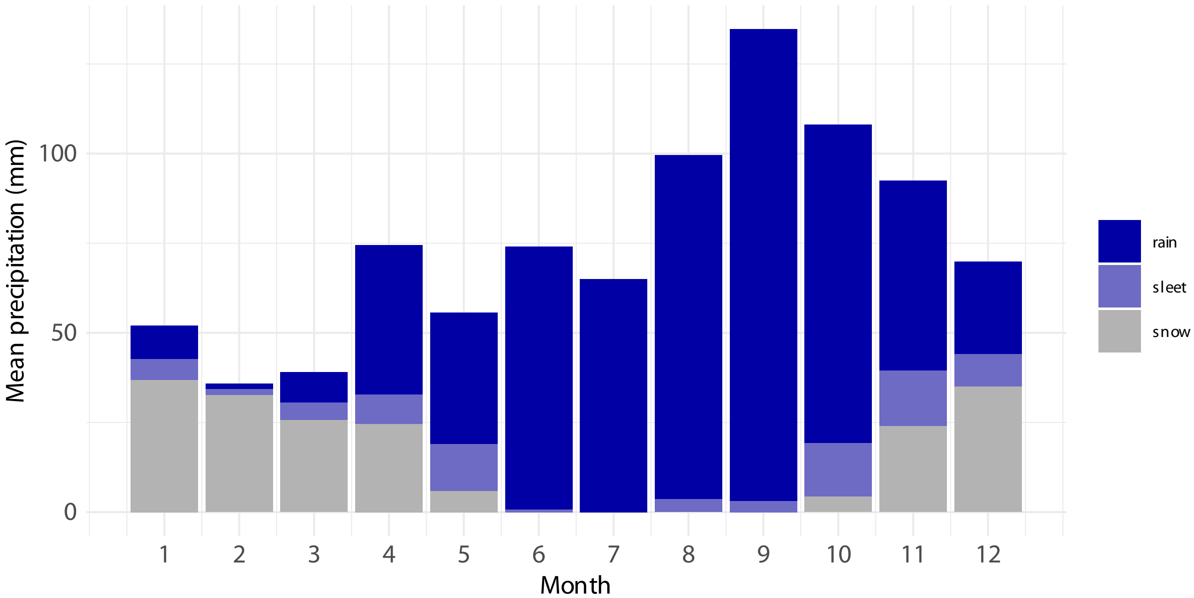

Figure 4: Mean precipitation type pr month. Mean monthly distribution of types of precipitation (mm). Based on data from 2008 to 2022 (requires join with temperature data which it only available till 2022). Sleet is defined as precipitation that fell when temperatures were between -1°C and 1°C.

Yearly mean precipitation

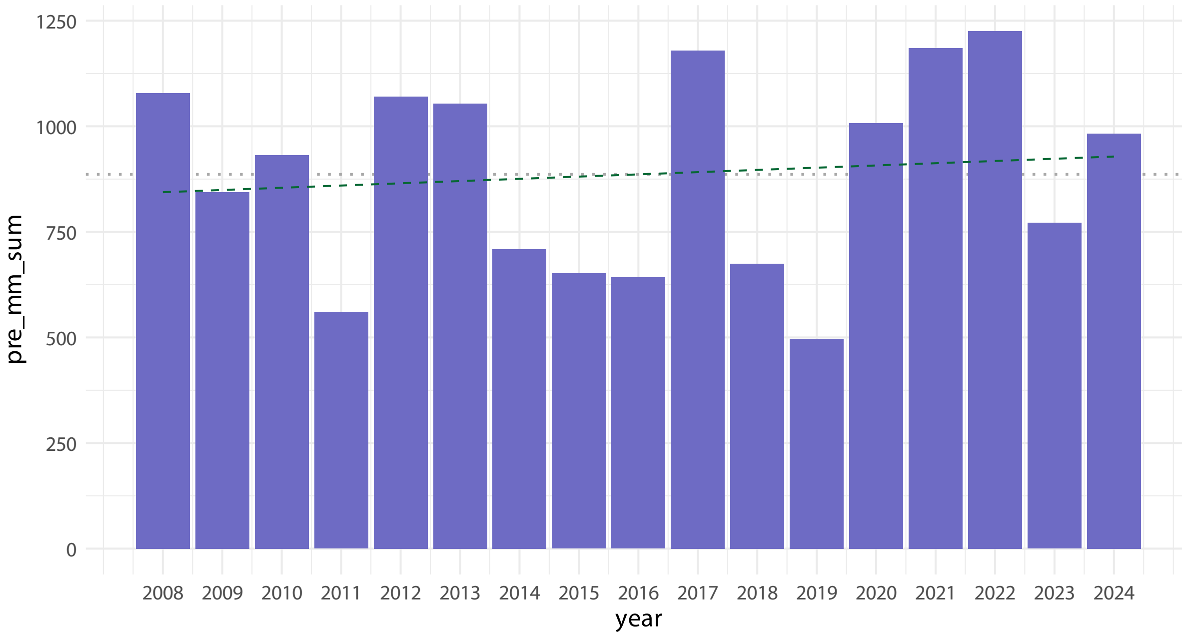

Figure 5: Mean yearly precipitation (mm) from 2008 to 2024. Data from 2007 is excluded because it was only from May - December. Blue dashed line indicate over all mean of 886.15 mm. Dashed line represent model prediction.

Five year interval precipitation

1-3 Radiation

Data available from https://data.g-e-m.dk (GEM 2026e)

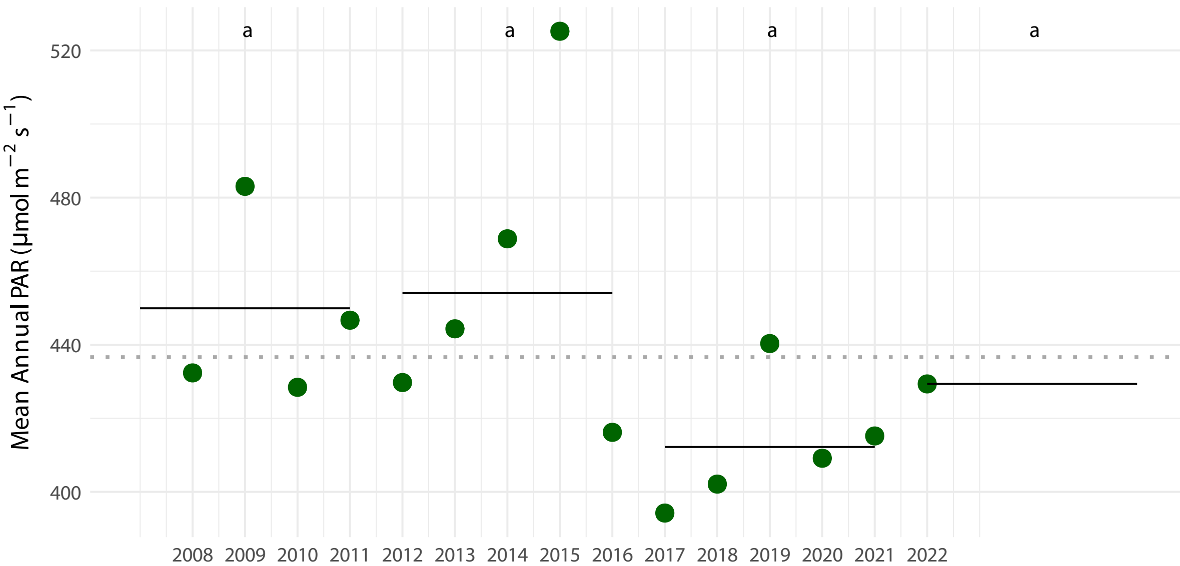

Figure 6: Five year intervals in PAR means. Shared letters indicate non-significant differences among estimated means according to compact letter display.

2 Historic climate data

2-1 Temperature

Data available from Cappelen (2021).

Station 4250 is Nuuk, and element 101 is precipitation.

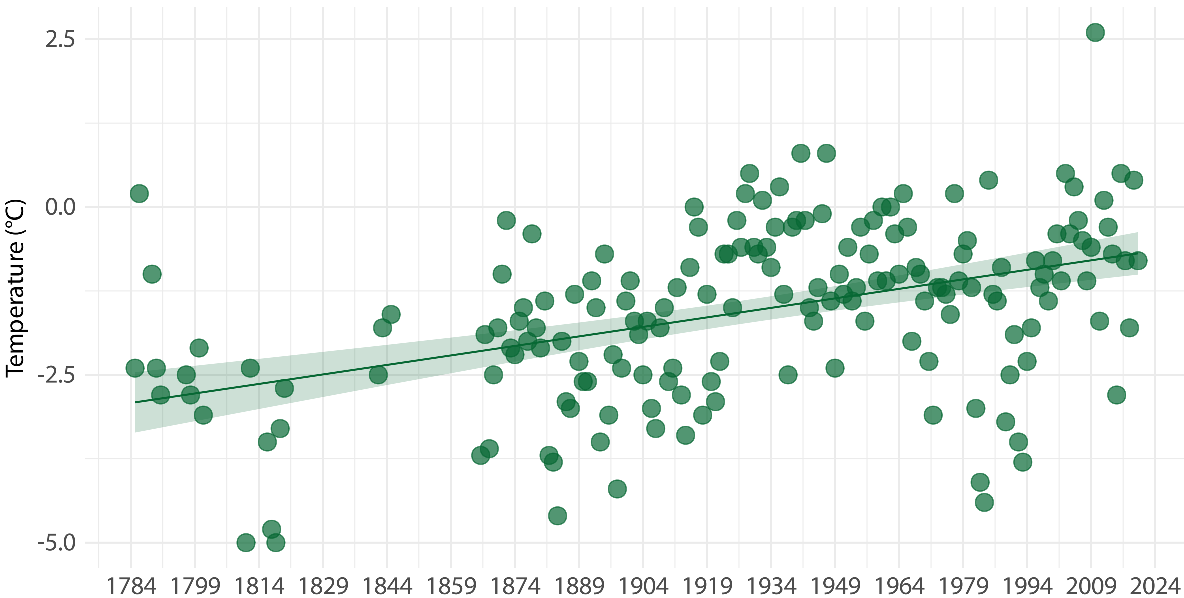

Figure 7: Historic yearly mean air temperature (°C) in Nuuk, from 1785 to 2020. Yearly mean of -1.6°C. Yearly mean ranges from -5°C to 2.6°C. Modelled mean temperature has increase from -2.92°C in 1784 to -0.65°C in 2024 (increase of 2.27°C or 0.095°C/decade.)

3 Statistical analysis

3-1 Diversity

Data available from https://data.g-e-m.dk (GEM 2026a).

Data has been filtered to remove sections with only one remaining plot and subsections in saltmarsh, due to too few replicates in this community type. Final calculates included 134 subsections (full transect consists of 162 subsection).

Richness

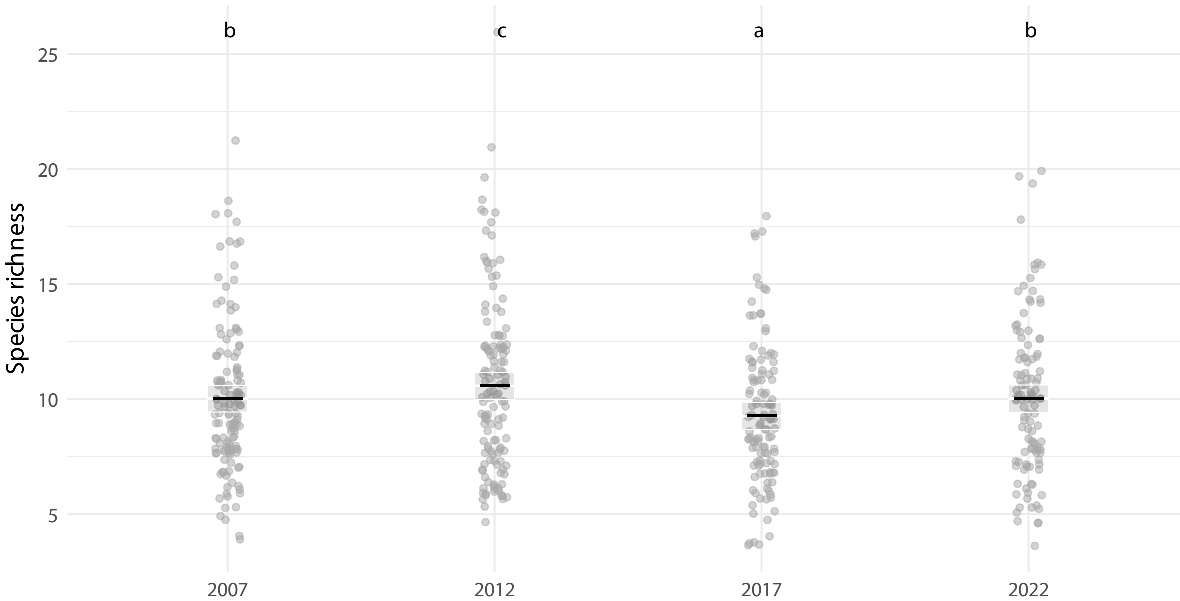

Figure 8: Species richness, number of species pr. subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Evenness

Figure 9: Evenness pr. subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Shannon

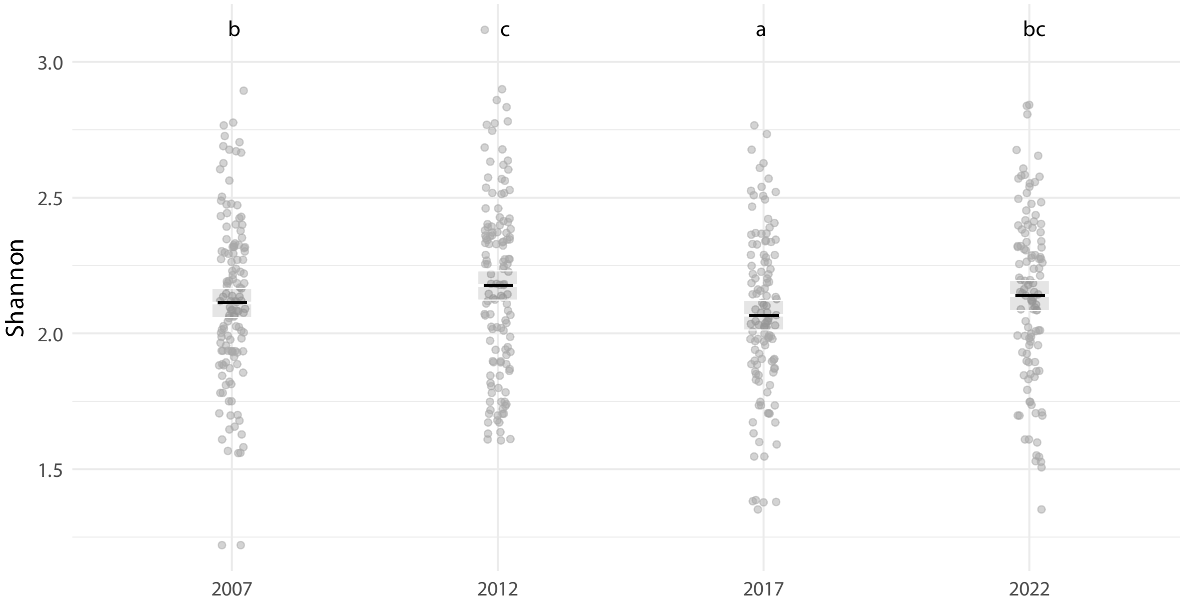

Figure 10: Shannon diversity pr. year calculate pr subsection. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Turnover

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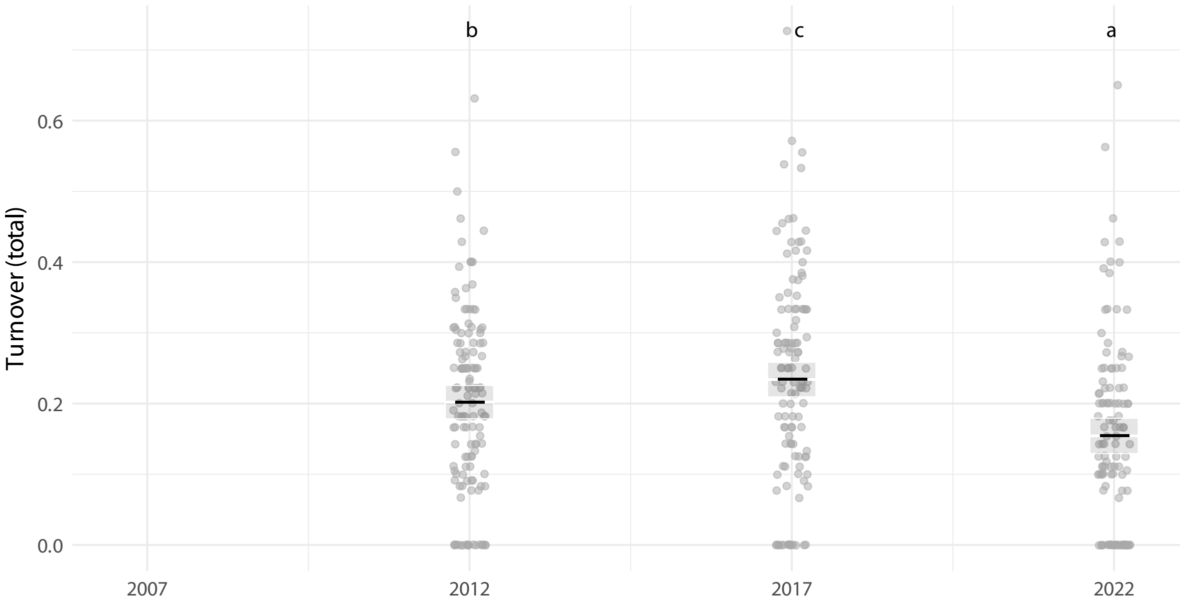

Figure 11: Turnover pr. year calculate pr subsection, but only pr present all survey years. A totalt of 131 subsections maintain more than 1 plots for all 4 surveys. Shared letters indicate non-significant differences among estimated means according to compact letter display.

Plots combined

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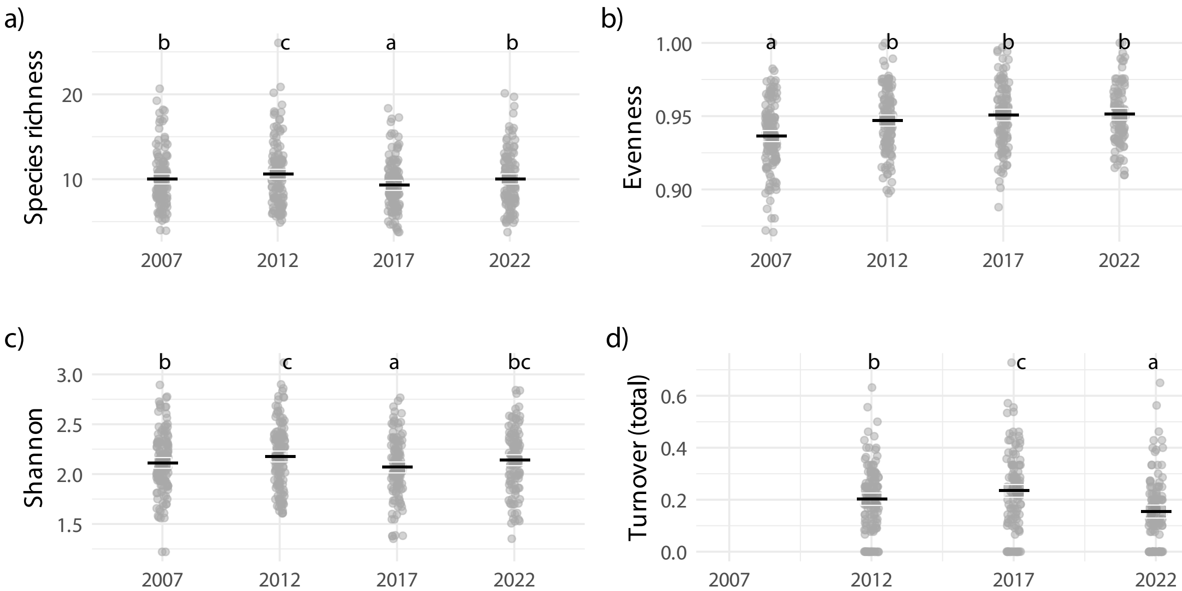


Figure 12: Diversity metrices, a) Shannon diversity, b) Species richness, c) Evenness, c) Species turnover. Shared letters indicate non-significant differences among estimated means according to compact letter display.

3-2 Community composition

NMDS

Community composition data ready for analysis consists of 134 subsections from 4 years of surveys. 99 species have been observed across all surveys.

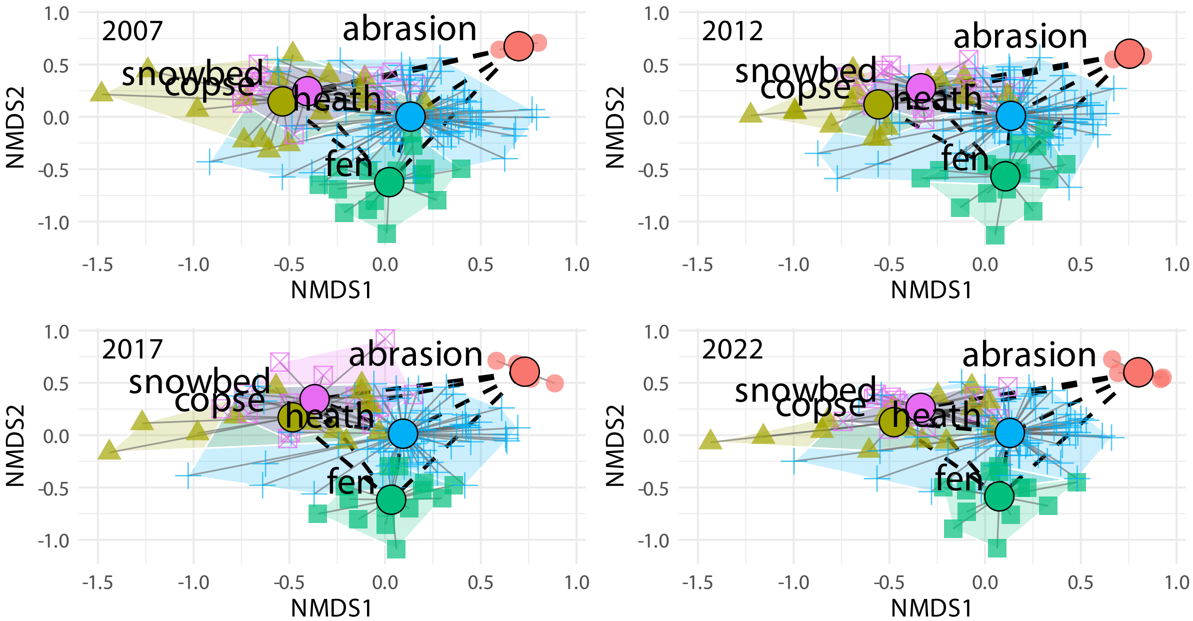


Figure 13: Plant community dynamics. Non-metric multidimensional scaling (NMDS) ordination of vegetation composition across years and vegetation types. Each panel represents subset of years of the full NMDS including all data to keep NMDS distances comparable. Grey lines connect plots within vegetation types, the within-type centroid distances. Black dashed lines indicate distances between vegetation-type centroids.

Between vegetation composition

```
Model: dist ~ factor(year) + (1 | veg_pair), fitted by REML

N observations: 40 | N groups (veg_pair): 10
```

Table 1: Fixed effects: centroid distance ~ survey year (factor)

term	estimate	std.error	statistic
(Intercept)	0.886	0.119	7.462
Year 2012	-0.014	0.017	-0.780
Year 2017	-0.018	0.017	-1.024
Year 2022	-0.025	0.017	-1.448

Table 2: Estimated mean centroid distances per survey

year	emmean	SE	lower.CL	upper.CL
2007	0.886	0.119	0.618	1.154
2012	0.873	0.119	0.605	1.140
2017	0.868	0.119	0.600	1.136
2022	0.861	0.119	0.593	1.129

Table 3: Pairwise contrasts between surveys (Sidak-adjusted)

contrast	estimate	SE	t.ratio	p.value
year2007 - year2012	0.014	0.017	0.780	0.970
year2007 - year2017	0.018	0.017	1.024	0.897
year2007 - year2022	0.025	0.017	1.448	0.647
year2012 - year2017	0.004	0.017	0.244	1.000
year2012 - year2022	0.012	0.017	0.668	0.986
year2017 - year2022	0.007	0.017	0.424	0.999

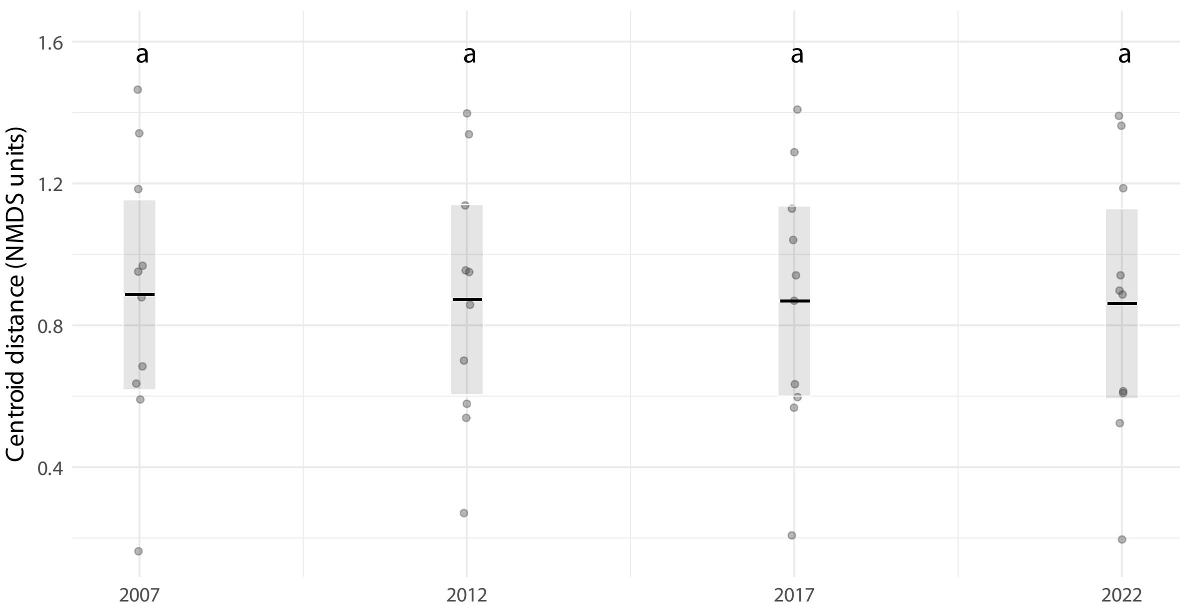


Figure 14: Between plant community dynamics. Estimated mean distance between vegetation-type centroids within each survey year based on linear mixed-effects models. Black lines are estimated means (± SE), and grey dots represent raw subsection-level distances (NMDS units). Shared letters indicate non-significant differences among estimated means according to compact letter display. (Figure 5)

Within vegetation composition

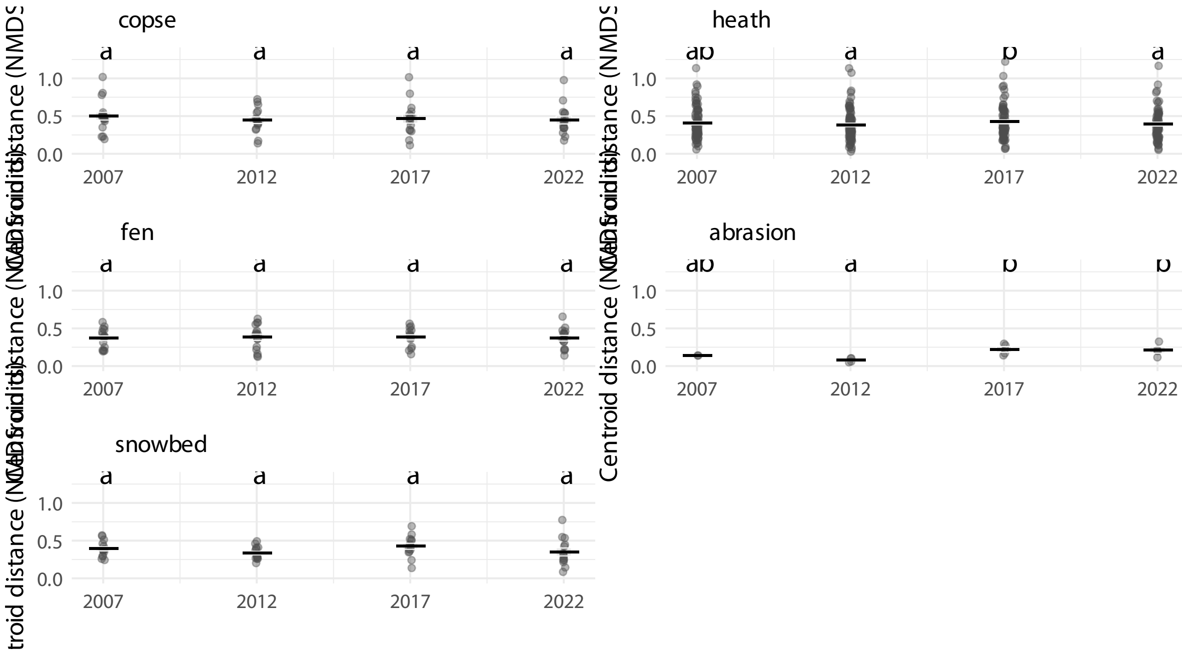


Figure 15: Within community dynamics. Estimated mean centroids-subsection distances pr community each survey year based on linear mixed-effects models. Black lines are estimated means (± SE), and grey dots represent raw distances (NMDS units). Shared letters indicate non-significant differences among estimated means according to compact letter display. (Figure 6)

3-4 Plant functional types

Functional groups

Table 4: Fixed effects per functional type

func_type	effect	term	estimate	std.error	statistic
forb	fixed	(Intercept)	0.269	0.029	9.253
forb	fixed	factor(year)2012	0.082	0.033	2.505
forb	fixed	factor(year)2017	0.004	0.034	0.113
forb	fixed	factor(year)2022	0.038	0.034	1.098
graminoid	fixed	(Intercept)	0.469	0.028	16.898
graminoid	fixed	factor(year)2012	0.078	0.038	2.022
graminoid	fixed	factor(year)2017	0.054	0.040	1.367

func_type	effect	term	estimate	std.error	statistic
graminoid	fixed	factor(year)2022	0.080	0.039	2.025
lichen	fixed	(Intercept)	0.491	0.035	14.114
lichen	fixed	factor(year)2012	0.141	0.021	6.780
lichen	fixed	factor(year)2017	-0.047	0.021	-2.225
lichen	fixed	factor(year)2022	0.050	0.021	2.326
shrub	fixed	(Intercept)	0.639	0.032	19.960
shrub	fixed	factor(year)2012	0.065	0.046	1.405
shrub	fixed	factor(year)2017	0.052	0.047	1.100
shrub	fixed	factor(year)2022	0.068	0.047	1.442
bryophyte	fixed	(Intercept)	0.886	0.016	55.617
bryophyte	fixed	factor(year)2012	0.062	0.016	3.955
bryophyte	fixed	factor(year)2017	-0.009	0.016	-0.599
bryophyte	fixed	factor(year)2022	0.062	0.016	3.885

Table 5: Estimated marginal means per functional type and survey year

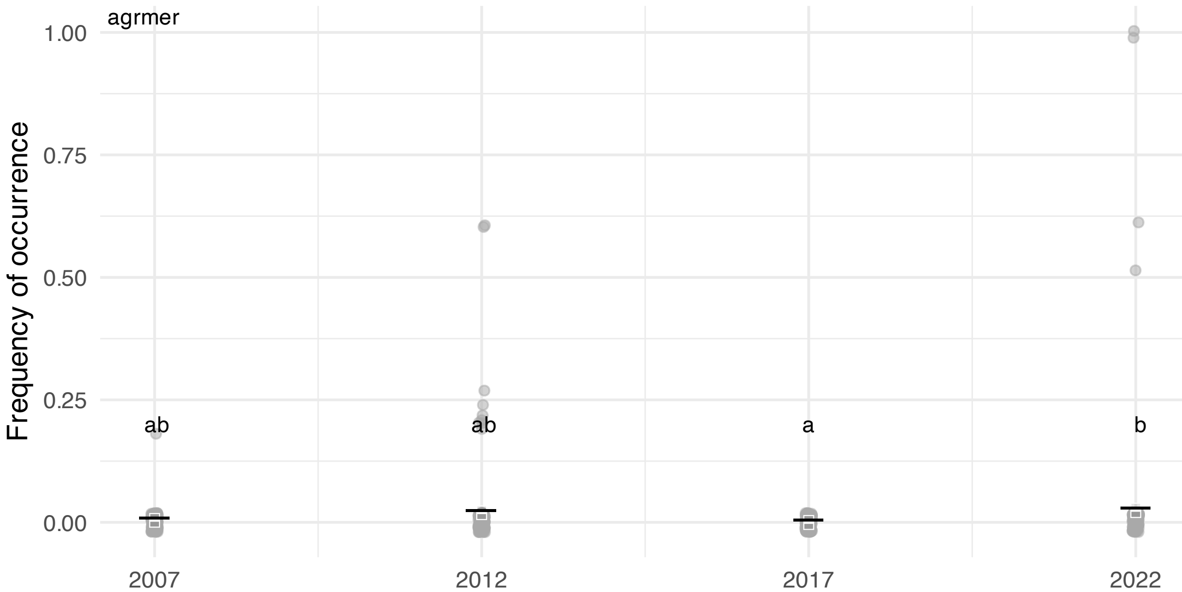
func_type	year	emmean	SE	df	lower.CL	upper.CL
forb	2007	0.269	0.029	464.397	0.211	0.326
forb	2012	0.351	0.028	430.029	0.295	0.407
forb	2017	0.272	0.030	499.890	0.214	0.331
forb	2022	0.306	0.030	502.004	0.247	0.366
graminoid	2007	0.469	0.028	924.733	0.415	0.524
graminoid	2012	0.547	0.028	947.164	0.491	0.603
graminoid	2017	0.523	0.030	1025.619	0.465	0.582
graminoid	2022	0.549	0.030	975.985	0.491	0.608
lichen	2007	0.491	0.035	184.632	0.422	0.560
lichen	2012	0.632	0.035	185.575	0.563	0.701
lichen	2017	0.444	0.035	188.852	0.375	0.513
lichen	2022	0.541	0.035	192.841	0.471	0.611
shrub	2007	0.639	0.032	1085.120	0.576	0.702
shrub	2012	0.703	0.033	1148.780	0.638	0.768
shrub	2017	0.691	0.035	1207.013	0.622	0.760
shrub	2022	0.707	0.035	1166.159	0.638	0.776
bryophyte	2007	0.886	0.016	302.899	0.854	0.917
bryophyte	2012	0.947	0.016	304.365	0.916	0.979
bryophyte	2017	0.876	0.016	313.374	0.844	0.908
bryophyte	2022	0.948	0.016	324.867	0.916	0.980

Table 6: Pairwise contrasts per functional type (Holm-adjusted)

func_type	contrast	estimate	SE	df	t.ratio	p.value
forb	year2007 - year2012	-0.082	0.033	1061.599	-2.501	0.075
forb	year2007 - year2017	-0.004	0.034	1069.739	-0.113	0.910
forb	year2007 - year2022	-0.038	0.034	1077.637	-1.096	0.819
forb	year2012 - year2017	0.078	0.034	1064.102	2.330	0.100
forb	year2012 - year2022	0.044	0.034	1072.424	1.315	0.756
forb	year2017 - year2022	-0.034	0.035	1056.859	-0.973	0.819
graminoid	year2007 - year2012	-0.078	0.038	1477.182	-2.020	0.260
graminoid	year2007 - year2017	-0.054	0.040	1493.905	-1.365	0.690
graminoid	year2007 - year2022	-0.080	0.039	1504.447	-2.022	0.260
graminoid	year2012 - year2017	0.024	0.040	1482.965	0.590	1.000
graminoid	year2012 - year2022	-0.002	0.040	1490.945	-0.056	1.000
graminoid	year2017 - year2022	-0.026	0.041	1472.875	-0.631	1.000
lichen	year2007 - year2012	-0.141	0.021	397.208	-6.754	0.000
lichen	year2007 - year2017	0.047	0.021	397.872	2.216	0.042
lichen	year2007 - year2022	-0.050	0.022	398.298	-2.317	0.042
lichen	year2012 - year2017	0.188	0.021	397.006	8.893	0.000
lichen	year2012 - year2022	0.091	0.021	397.183	4.255	0.000
lichen	year2017 - year2022	-0.097	0.022	396.530	-4.491	0.000
shrub	year2007 - year2012	-0.065	0.046	1719.484	-1.403	0.900
shrub	year2007 - year2017	-0.052	0.047	1749.981	-1.098	1.000
shrub	year2007 - year2022	-0.068	0.048	1768.947	-1.440	0.900
shrub	year2012 - year2017	0.013	0.048	1736.864	0.260	1.000
shrub	year2012 - year2022	-0.004	0.048	1754.299	-0.079	1.000
shrub	year2017 - year2022	-0.016	0.050	1719.352	-0.330	1.000
bryophyte	year2007 - year2012	-0.062	0.016	399.038	-3.940	0.000
bryophyte	year2007 - year2017	0.009	0.016	400.994	0.597	1.000
bryophyte	year2007 - year2022	-0.062	0.016	402.436	-3.870	0.000
bryophyte	year2012 - year2017	0.071	0.016	399.165	4.489	0.000
bryophyte	year2012 - year2022	-0.001	0.016	399.852	-0.038	1.000
bryophyte	year2017 - year2022	-0.072	0.016	397.946	-4.440	0.000

Individual species

A total of 97 species were analysed. 18 of them had significantly different means between survey years. Plots of the frequency of occurrence for these 18 species are rendered below.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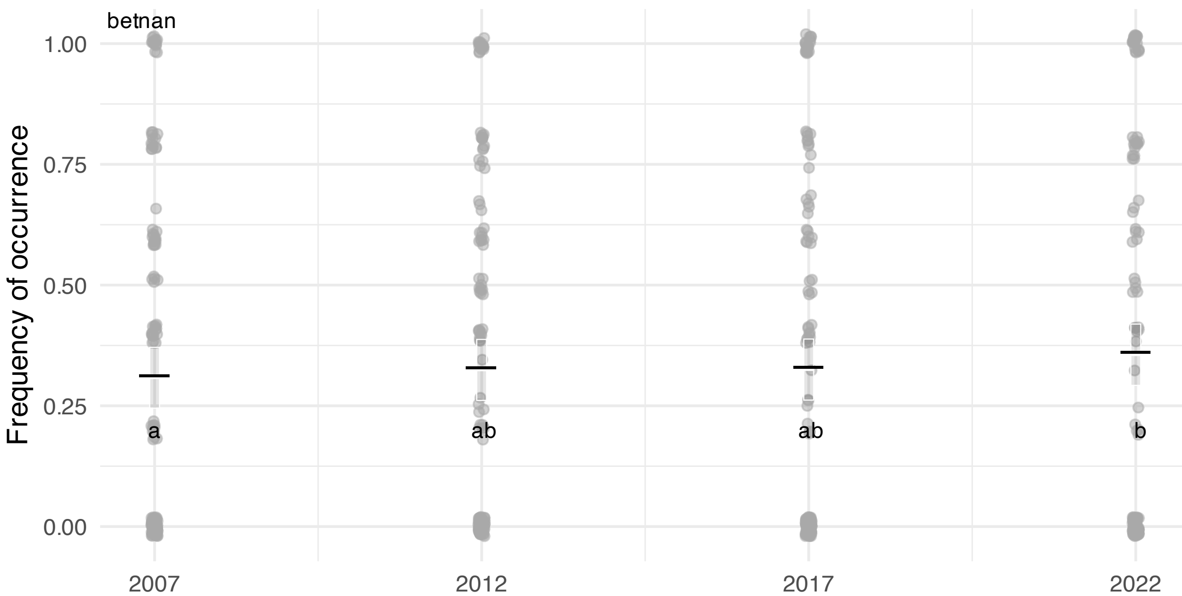
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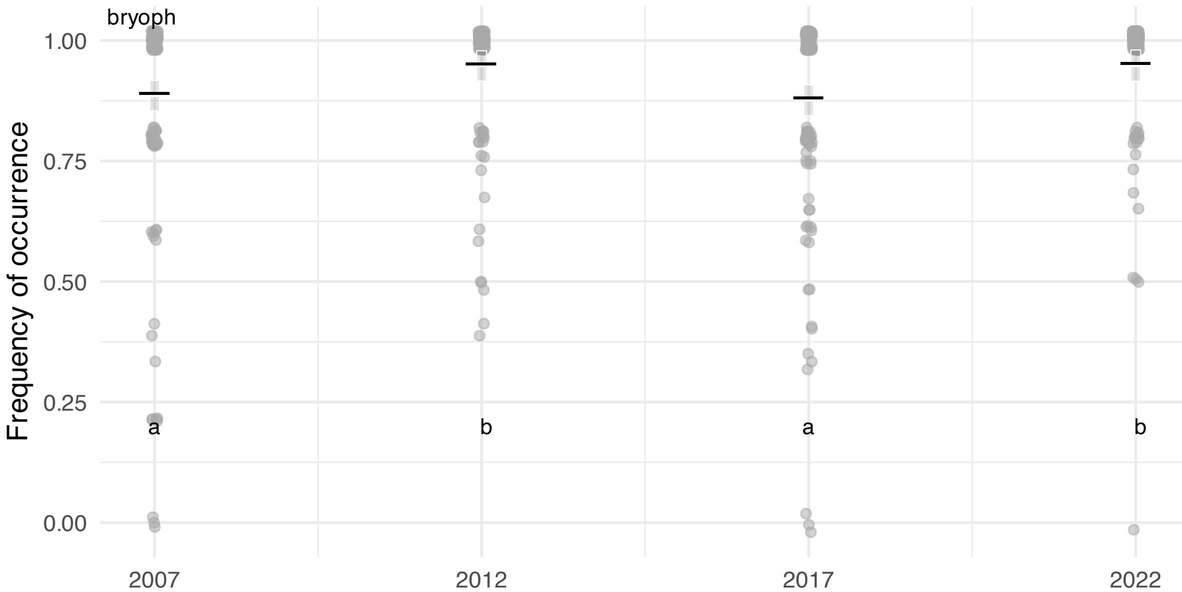
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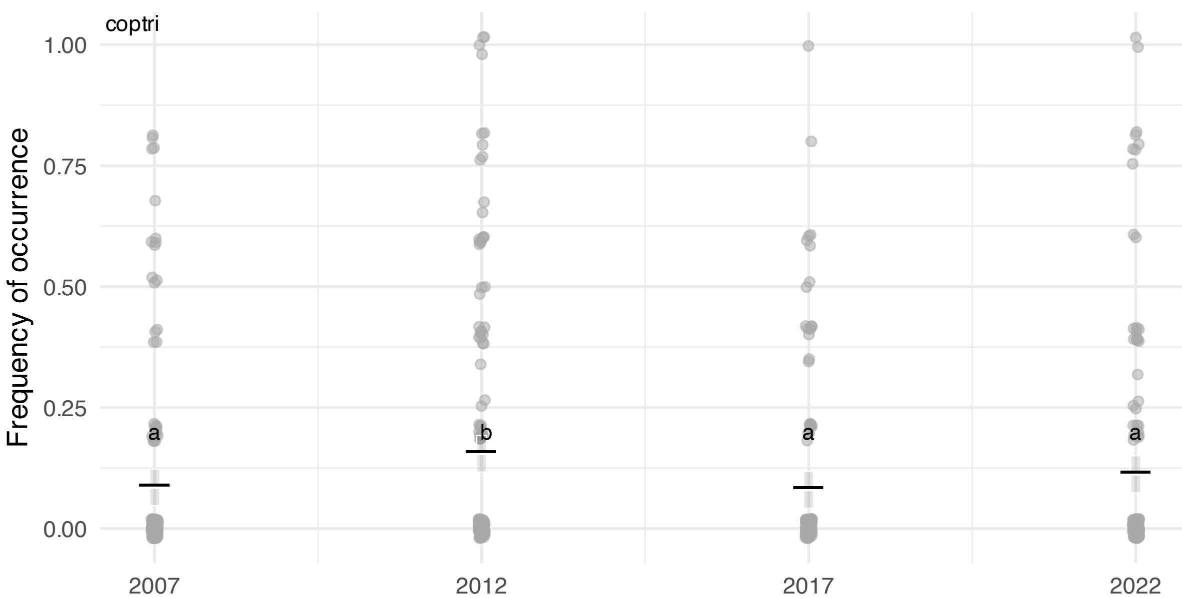
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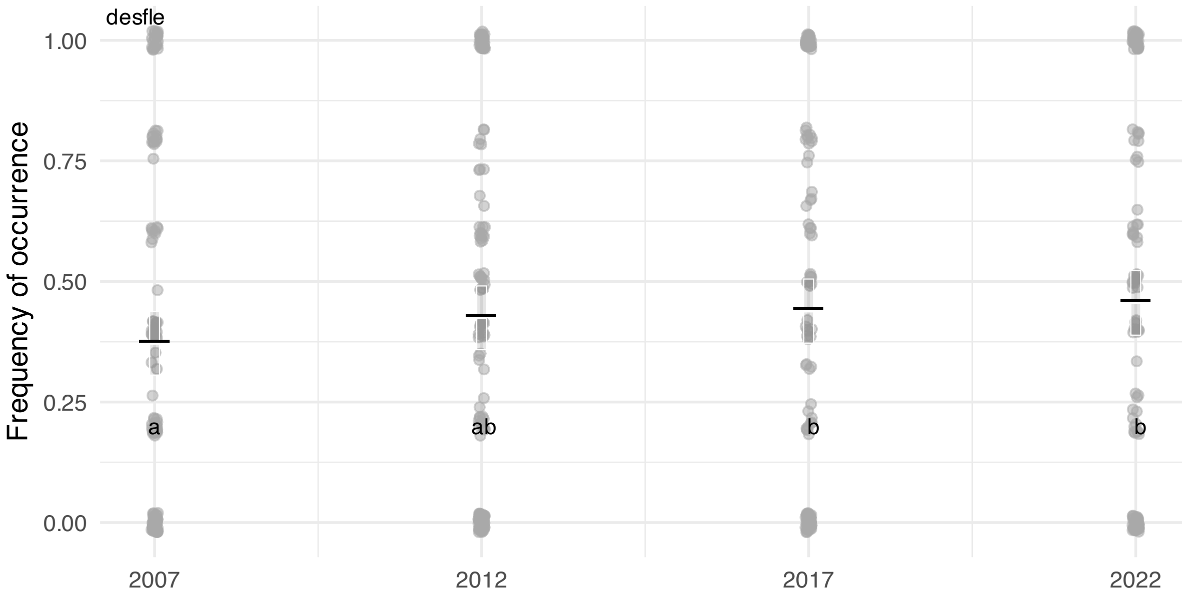
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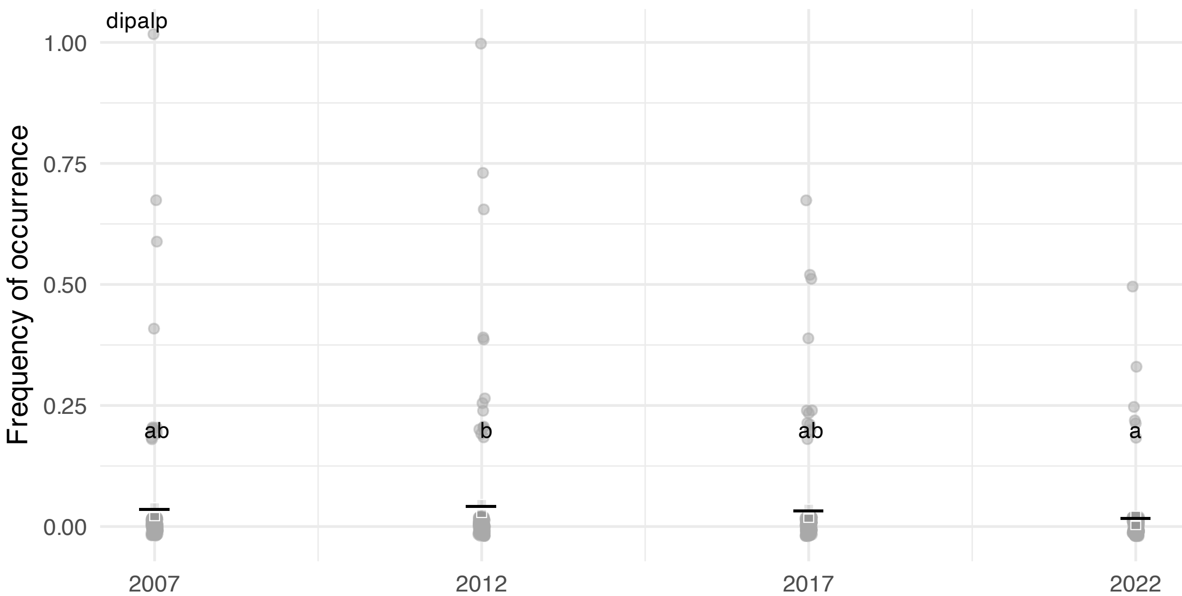
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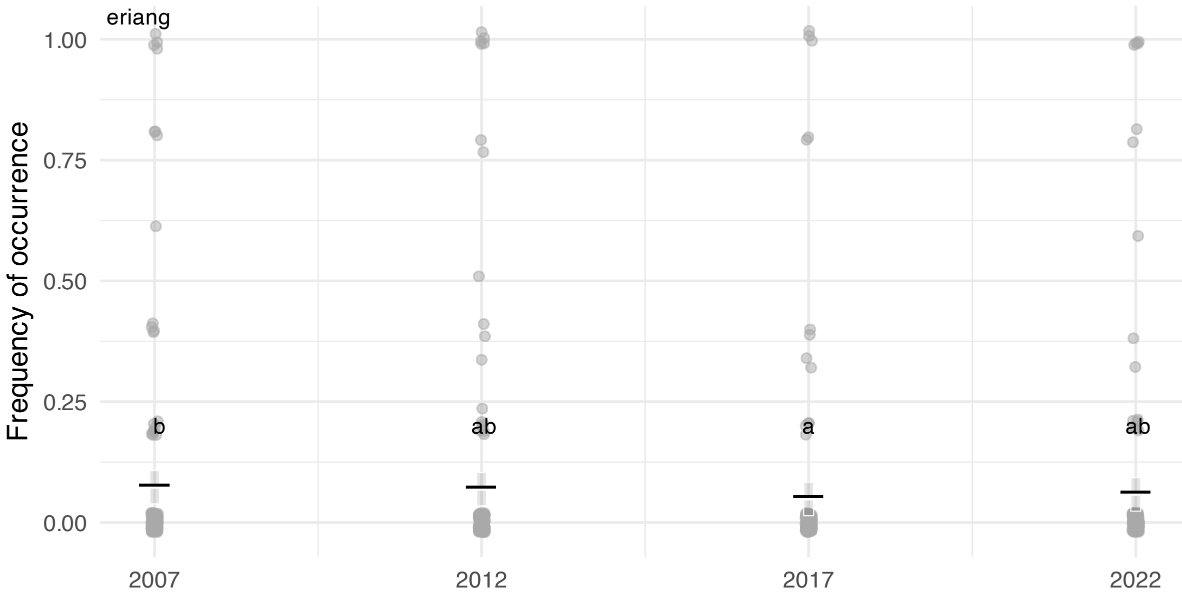
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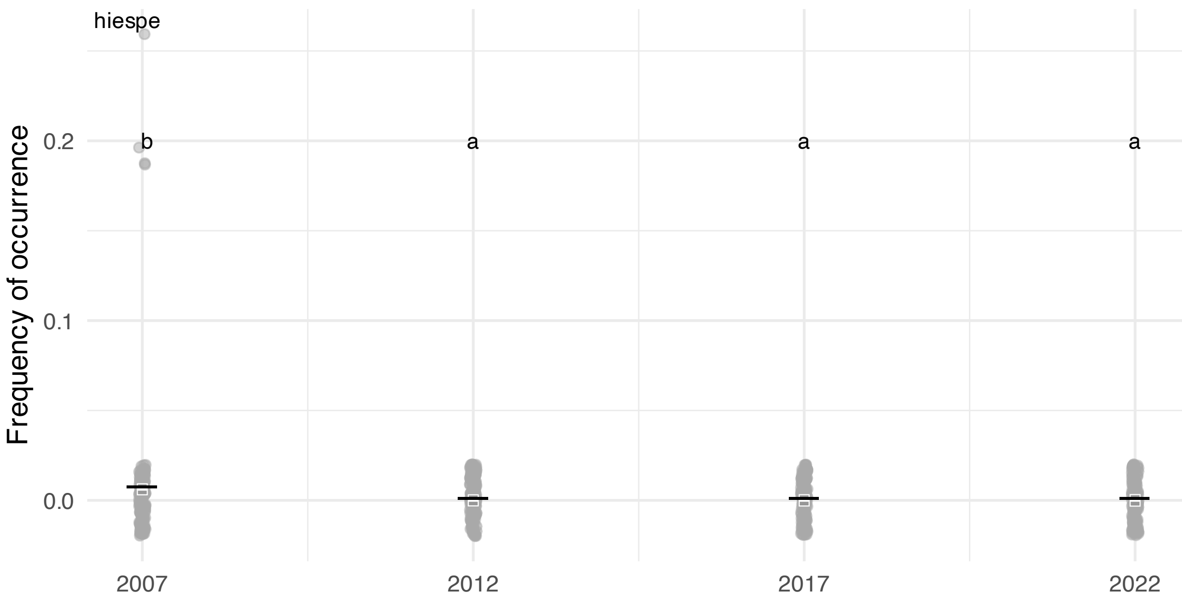
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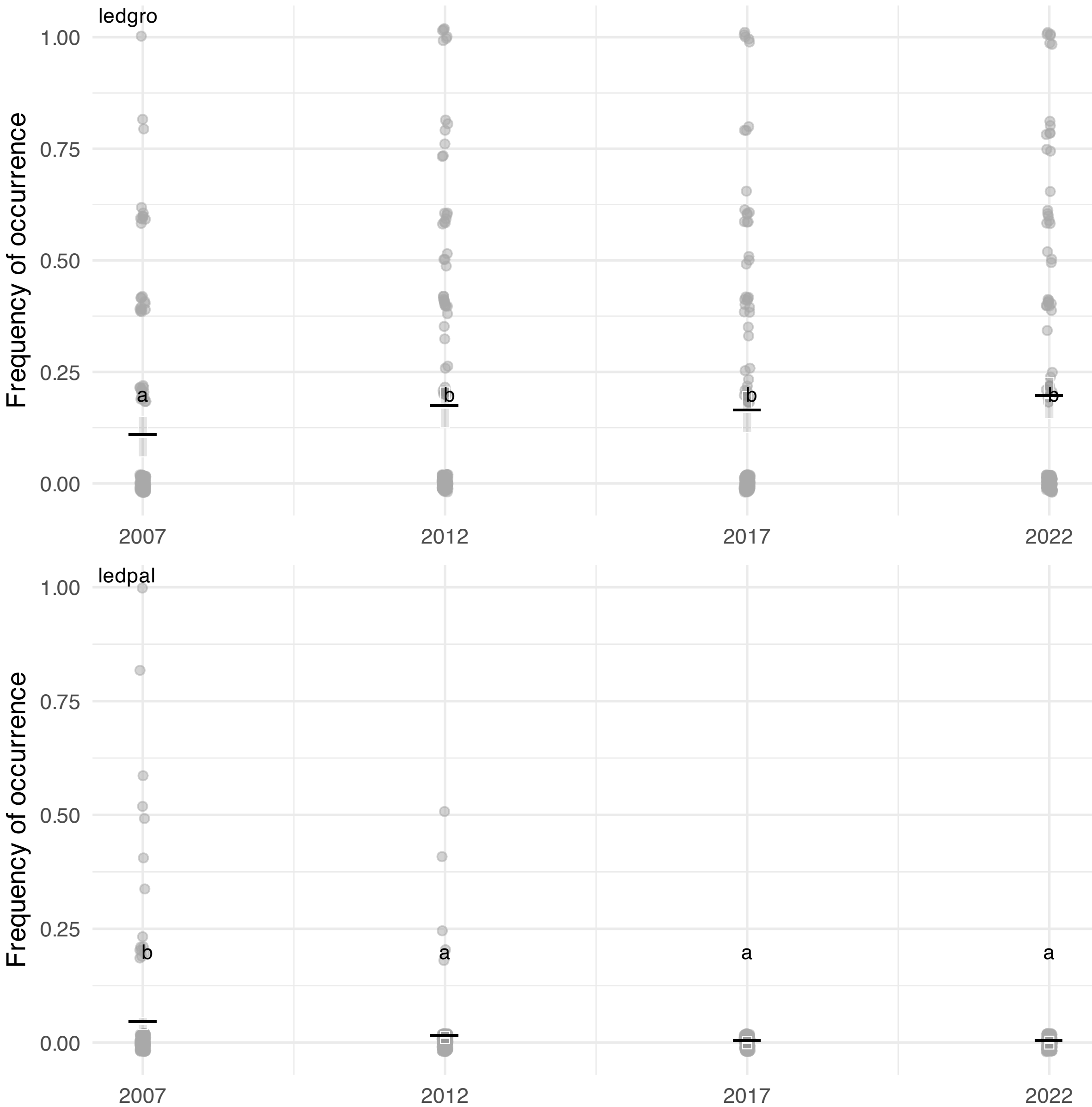
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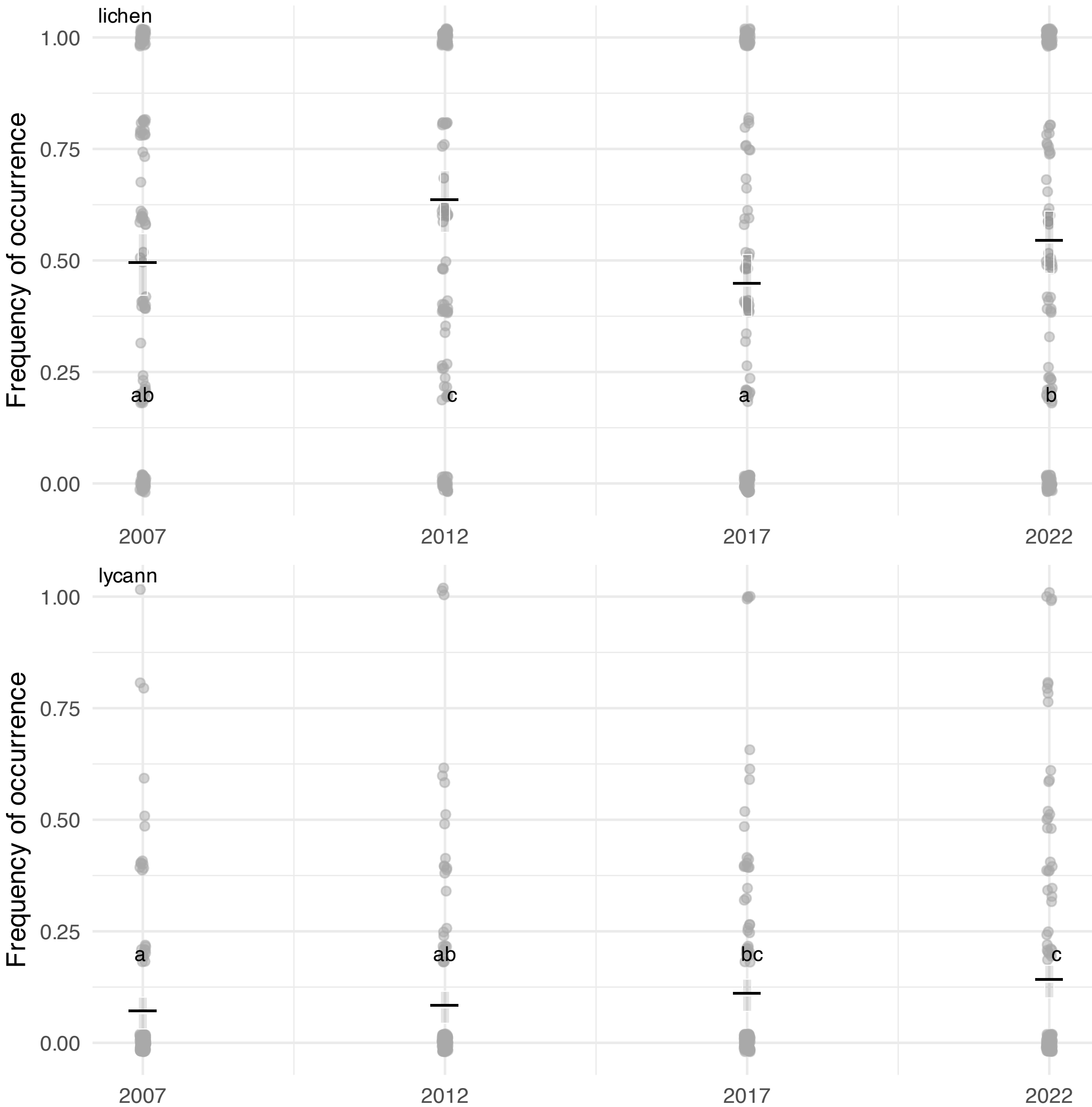
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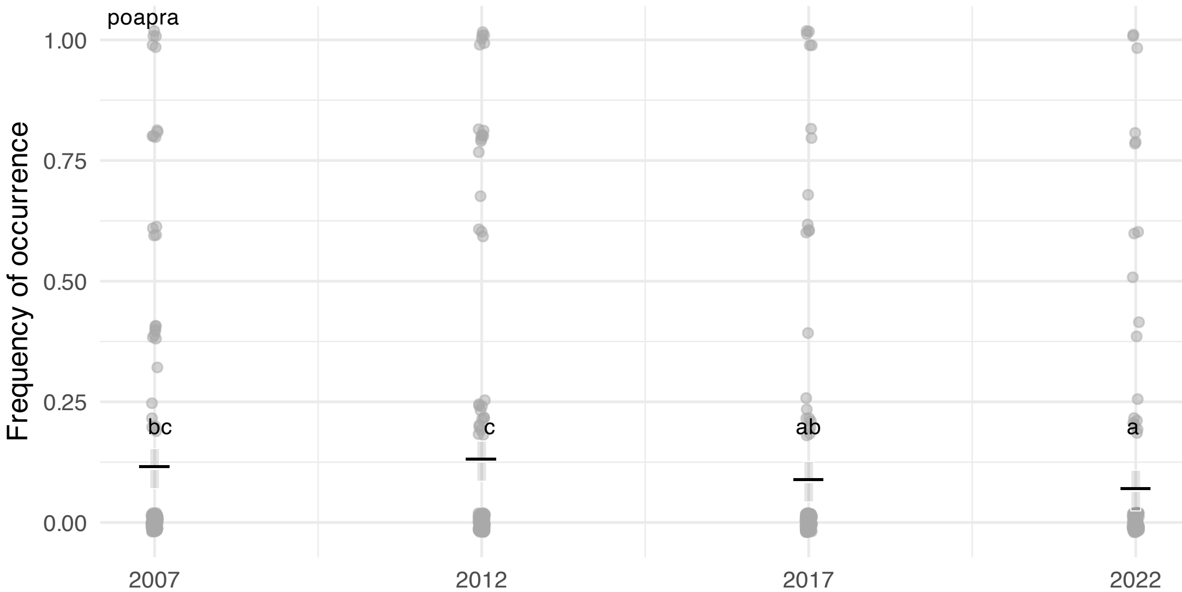
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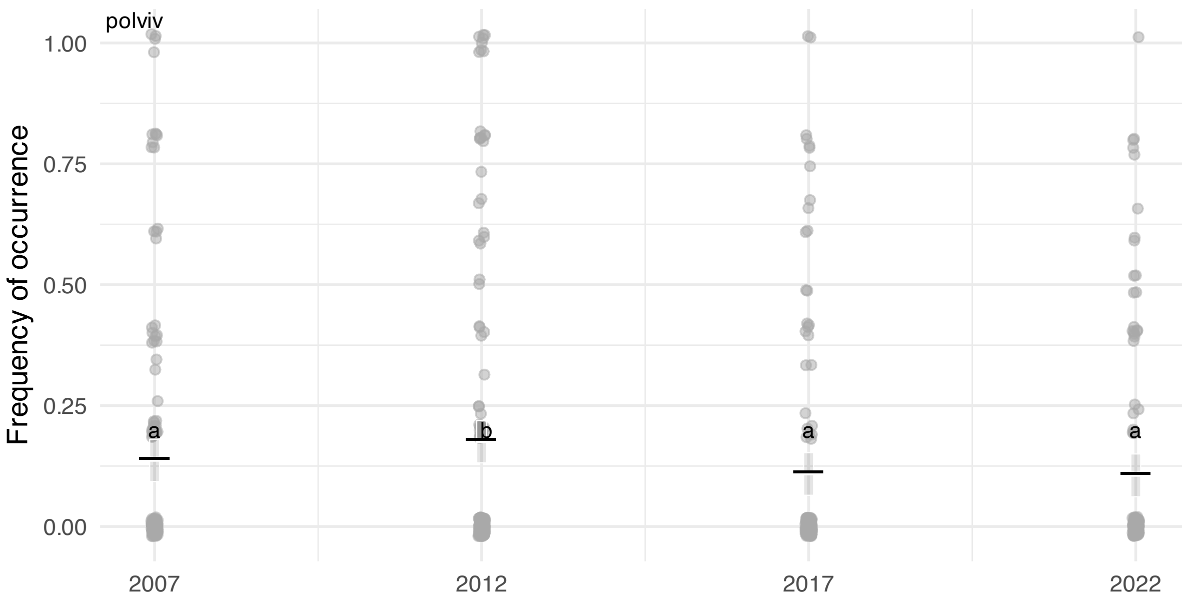
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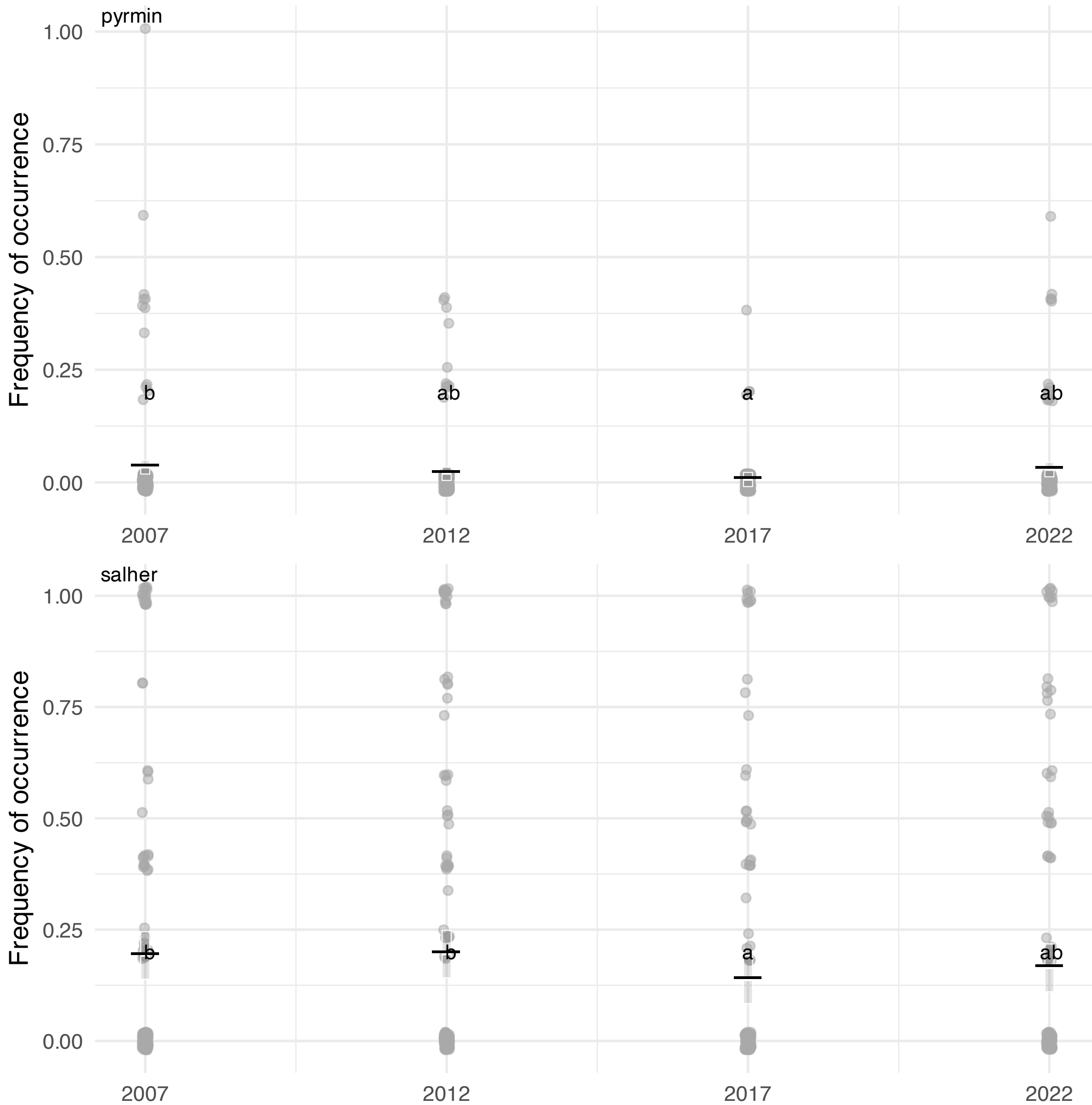
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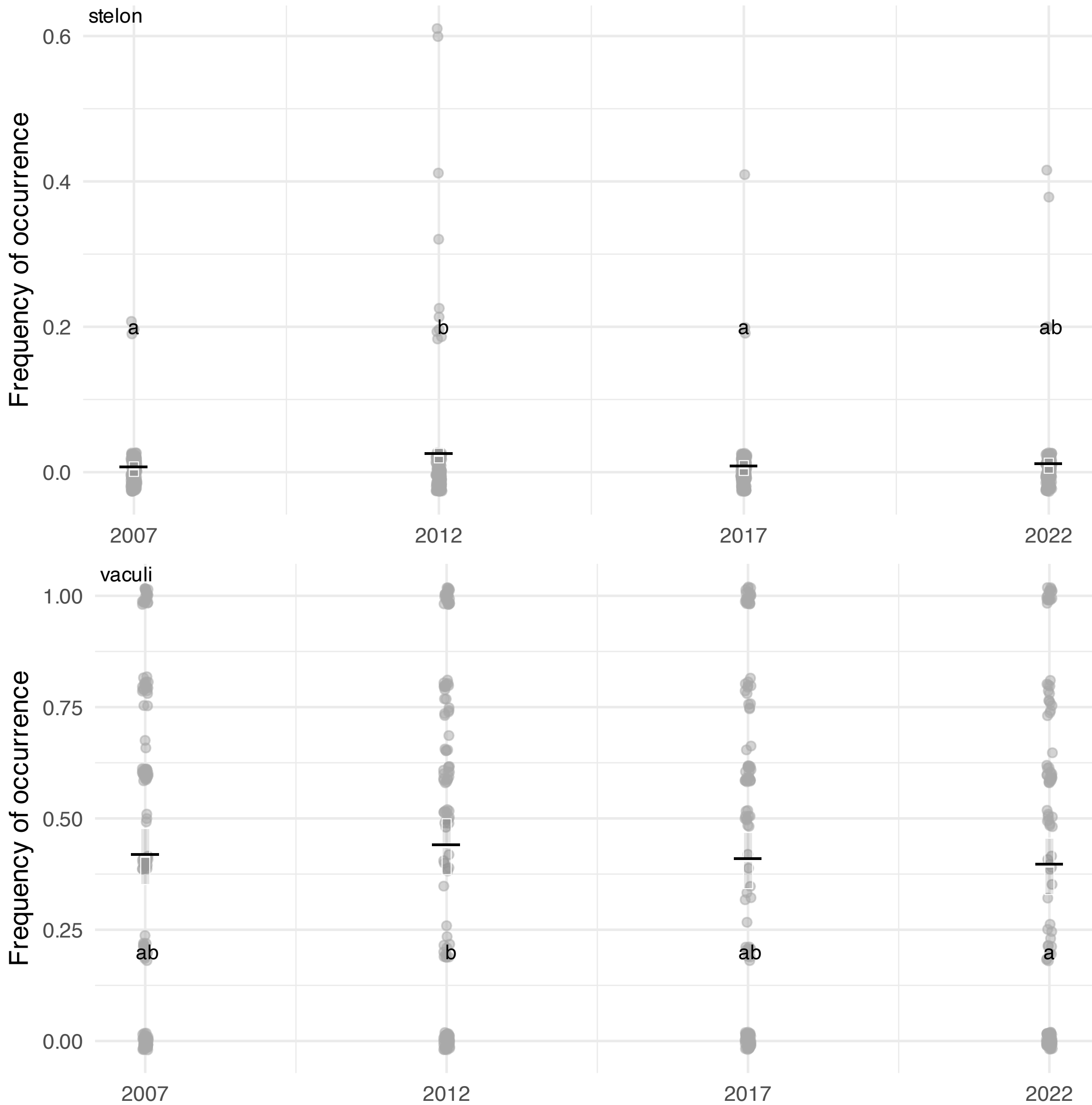

4 Phenology (discussion)

Data available from https://data.g-e-m.dk (GEM 2026b, 2026c).

4-1 NDVI

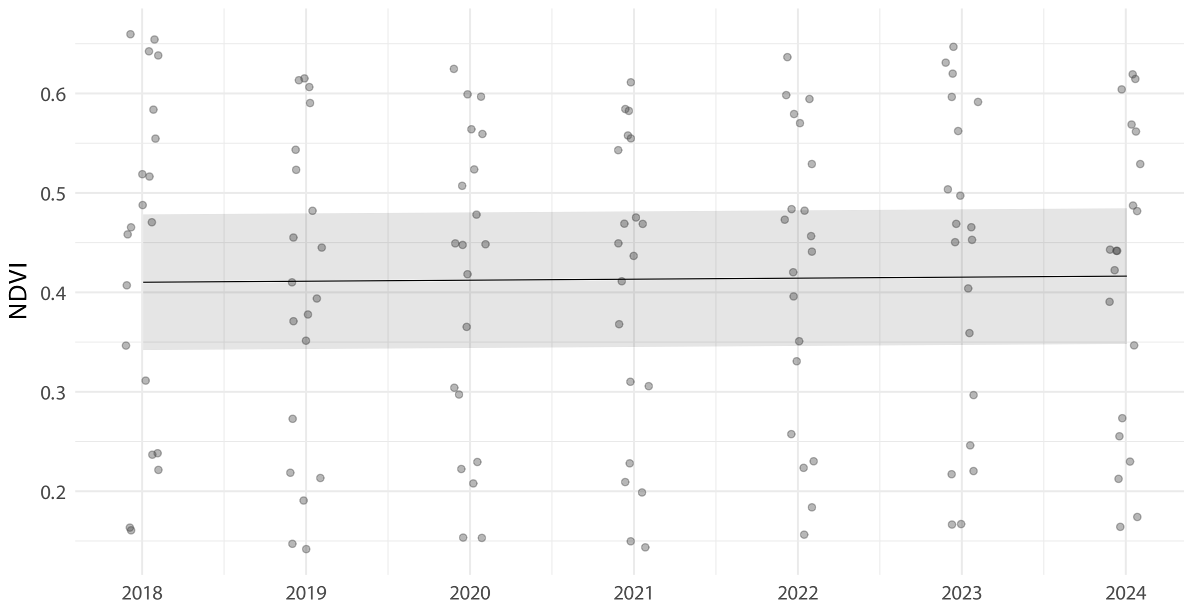

Figure 16: NDVI change over time. Mean NDVI per plot per year, based on in situ measurements from 2018 to 2024. The trend line shows a significant increase of 0.001 NDVI units per year (95% CI: 3e-04–0.0018), representing a cumulative increase of 0.0061 NDVI units over the study period.

5 References

Cappelen, John. 2021. *Greenland – DMI Historical Climate Data Collection 1784-2020*. Nos. 21–04. Danish Meteorological Institute. https://www.dmi.dk/fileadmin/Rapporter/2021/DMIRep21-04.pdf.

GEM. 2026a. "BioBasis Nuuk - Vegetation - NERO Line Raunkiaer." Version 1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/HXJY-RQ13.

GEM. 2026b. "BioBasis Nuuk - Vegetation - Plot NDVI." Version 1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/TQTT-EV69.

GEM. 2026c. "BioBasis Nuuk - Vegetation - Total Count (Eriophorum Angustifolium, Loiseleuria Procumbens, Salix Glauca, Silene Acaulis)." Version 1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/J6EK-FR47.

GEM. 2026d. "ClimateBasis Nuuk - Precipitation - Precipitation Accumulated (Mm)." Version

1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/SXJ8-WA79.

GEM. 2026e. "ClimateBasis Nuuk - Radiation - Photosyntetic Active Radiation @ 200 Cm - 5min Average (Mmol/M2/Sec)." Version 1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/8Z2W-D993.

GEM. 2026f. "ClimateBasis Nuuk - Temperature - Air Temperature @ 200 Cm - 30min Average (°C)." Version 1.0. Greenland Ecosystem Monitoring. https://doi.org/10.17897/PGN3-7597.